

Restaurant Management System

Software Documentation

Developed By: Syed Saad Raza
Programming Language: C++ with Qt
Database: MySQL

1. Introduction

The Restaurant Management System is a desktop application developed using C++ and the Qt framework. The purpose of this system is to automate restaurant operations such as order management, menu handling, staff management, and sales tracking. Traditionally, many restaurants record transactions and orders manually on paper, which can lead to errors and data loss. This system digitizes those processes, allowing data to be stored securely in a MySQL database for long-term access.

2. Objectives of the System

- To automate restaurant operations.
- To maintain digital records of sales, orders, and menu items.
- To reduce manual work and human error.
- To track order progress from kitchen to serving staff.
- To provide profit and sales insights to restaurant owners.

3. System Users

The system is designed for multiple types of users within the restaurant environment:

- Admin – Manages staff, menu items, and tables.
- Order Taker – Handles dine-in and takeaway orders.
- Owner – Views reports, staff details, and sales profits.
- Kitchen Staff – Updates order status such as In Progress or Ready.

4. Functional Requirements

The system provides the following functionalities:

- Secure login system with role-based access.
- Staff management including add, update, and delete operations.
- Menu management for adding and modifying food items.
- Table management for dine-in customers.
- Order management supporting takeaway and dine-in orders.
- Kitchen order tracking system.
- Automatic notifications when orders are ready.
- Bill generation and printing.
- Sales reporting and profit calculation.

5. Non-Functional Requirements

- The system should be user-friendly and easy to operate.
- The system should provide fast database queries.
- The system should store data securely in MySQL.
- The system should support multiple user roles.
- The application should run smoothly on Windows OS.

6. System Architecture

The system follows a Two-Tier Client-Server Architecture.

Client Layer: Qt Desktop Application developed using C++.

Data Layer: MySQL Database which stores restaurant data.

The client application communicates directly with the database to insert, update, and retrieve restaurant information.

7. System Modules

7.1 Login Module

Provides authentication and role-based access control for Admin, Order Taker, Owner, and Kitchen staff.

7.2 Staff Management Module

Allows administrators to add, update, or remove staff members and manage salary and positions.

7.3 Menu Management Module

Allows administrators to manage food items, categories, cost price, selling price, and images.

7.4 Order Management Module

Handles both takeaway and dine-in orders. Order takers can add items to orders and generate bills.

7.5 Kitchen Tracking Module

Kitchen staff can mark orders as In Progress or Ready, allowing the system to notify the order taker.

7.6 Sales Reporting Module

The owner can view sales reports and profit analysis based on stored order data.

8. Database Design

The system uses MySQL as its backend database. Important tables include:

- staff – Stores employee details.
- menu – Stores food items and categories.
- orders – Stores overall order information.
- order_items – Stores individual items inside each order.
- restaurant_tables – Stores dine-in table numbers.

9. Order Processing Data Flow

1. Order taker selects food items from the system.
2. The order is stored in the MySQL database.
3. Kitchen staff views the order in the kitchen dashboard.
4. Kitchen staff marks the order as In Progress or Ready.
5. The system notifies the order taker when the order is ready.
6. Order taker prints the bill and marks the order as paid.

10. Installation Guide

1. Install Qt Creator IDE.
2. Install MySQL Server and create the restaurant database.
3. Import the database schema.
4. Configure MySQL connector in the Qt project.
5. Run the application using Qt Creator.

11. Testing

The system was tested for the following scenarios:

- Login validation testing.
- Order placement testing.
- Kitchen order status updates.
- Database insertion and retrieval testing.
- Bill generation and printing functionality.

12. Conclusion

The Restaurant Management System successfully automates restaurant operations including order management, menu management, and sales reporting. The system reduces manual work and improves efficiency while providing valuable business insights to restaurant owners.